Exploration of Machine Learning Models for Fraud Detection

TEAM -1

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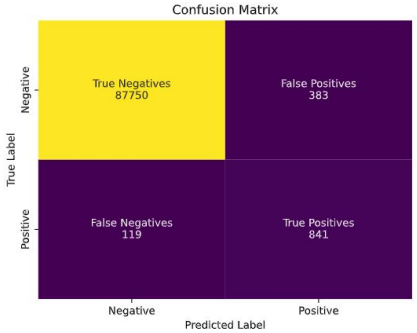
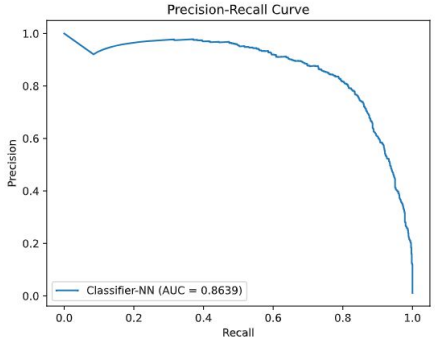
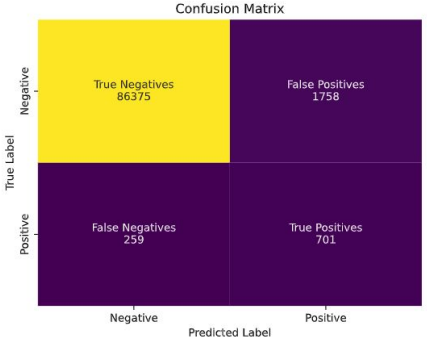
Model 1: Classifier Neural Network

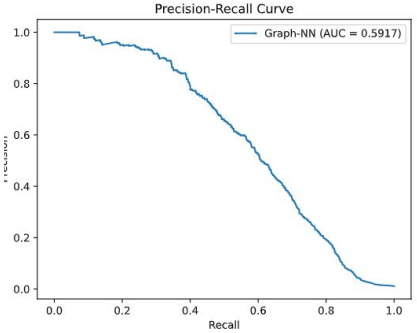




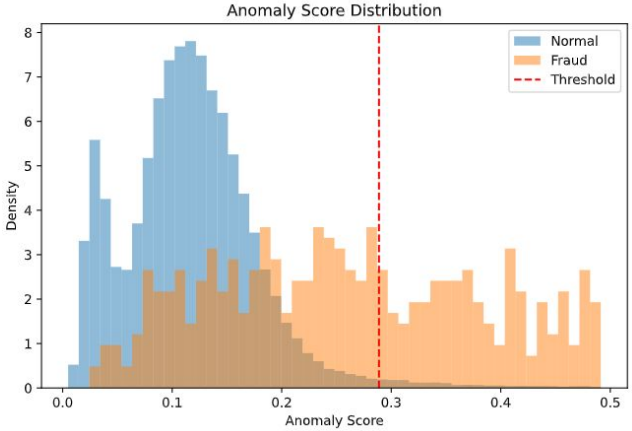
The poorer performance is likely because our features already contained strong signals (e.g., transaction patterns). Tree models can immediately exploit these, whereas neural nets need to learn representations from scratch. **Adding more depth to the net in future may improve the model's pattern detection**.

Model 2: Graph Neural Network

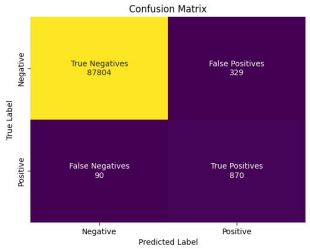


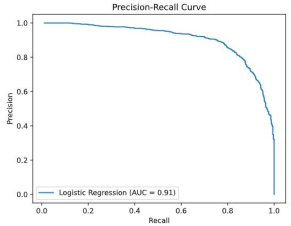


Shows the clustering density relationship between fraud and non-fraud transactions. **We suspect that the model's poor performance can be attributed to the synthetic data in which fraud transactions had very similar behaviour to non-fraud cases, as seen in the overlap above**. This made it difficult for the model to differentiate and cluster accurately.



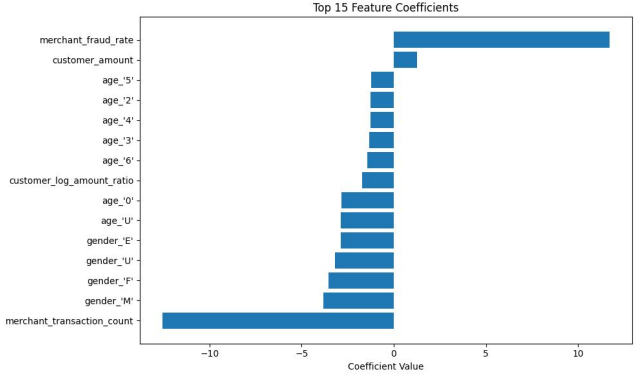
Model 3: Logistic Regression



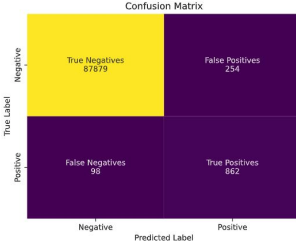


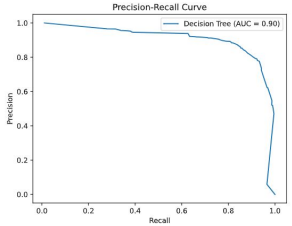
The curve comes relatively close to the top-right corner, with a PR score of 0.91. The model was very successful in identifying the majority of fraud cases in our test dataset.

Merchant fraud rate is the most influential feature for this model, followed by the average transaction amount of that customer. We see that age and gender both have very low (negative) importance, which is evidence of an unbiased and reasonable decision-making process.



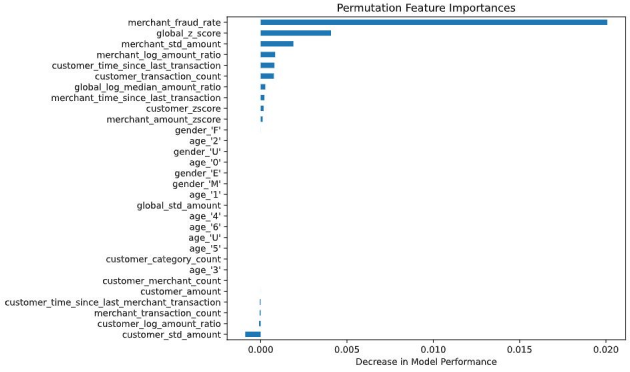
Model 4: Decision Tree



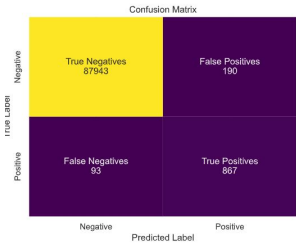


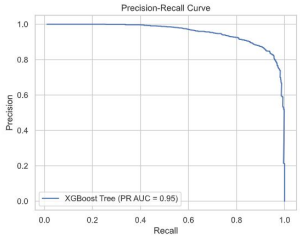
Like Logistic Regression, focuses mainly on merchant fraud rate which shows good decision making. **However interestingly uses more of the merchant's attributes than customer's.**

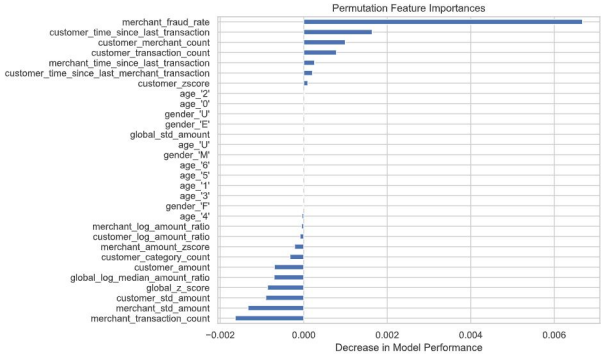
Our feature engineering ensured very high precision and recall. However, we finetuned the model using a random 15 hyperparameter samples. In future we could use all 150 sample hyperparameters to find the absolute optimal combination.



Model 5: XGBoost





Most influenced by merchant fraud rate again, however a stronger focus on customer attributes than the Decision Tree. **This may be a more reasonable choice as it accounts for customer's past transaction amounts and merchant history**.

XGBoost uses gradients in backpropagation to minimise loss, finding the best possible tree. It also handles noisy data quite well, and worked around the ambiguity in synthetic data behaviours which impacted the graph neural net. This makes it our best, most accurate model, evident in the accuracy table in our report.